

X Решения II уровня

Т 10.1

Дано:

$$m = 306 \text{ г.}$$

$$l = 120 \text{ см}$$

$$\Delta \alpha = 10^\circ$$

Найти: M_{TP}

Решение:

$$\vec{M}_{TP} = [\vec{r}_1 \times \vec{L}]$$

$$\vec{M}_{mg} = [\vec{r}_n \times \vec{L}] \quad M_{mg} = mgl$$

$$\frac{M_{TP}}{mgl} = \frac{\Delta \alpha}{360^\circ}$$

$$M_{TP} = mgl \cdot \frac{1}{36} = 0,01 \text{ Н.м}$$

Ответ: 0,01 Н.м

Т 10.2

Дано:

$$m; x; M_{TP}$$

Найти: N

Решение:

$$\frac{dL}{dt} = M_{TP}$$

$$M = mgx = \Omega \cdot L = \frac{d\varphi}{dt} L$$

$$mgx = \frac{d\varphi}{dt} \cdot M_{TP} L$$

$$\frac{mgx}{M_{TP} L} dL = d\varphi$$

$$\frac{mgx}{M_{TP}} \ln \frac{L_k}{L_n} = -\varphi$$

$$\frac{mgx}{M_{TP}} \ln \frac{I_0 \omega_n}{I_0 \omega_k} = \varphi \quad \frac{\omega_n}{\omega_k} = 2$$

$$N = \frac{\varphi}{2\pi} = \frac{mgx \ln 2}{2\pi M_{TP}}$$

Ответ: $\frac{mgx \ln 2}{2\pi M_{TP}}$

Т 10.3

Дано:

$$m = 200 \text{ кг}$$

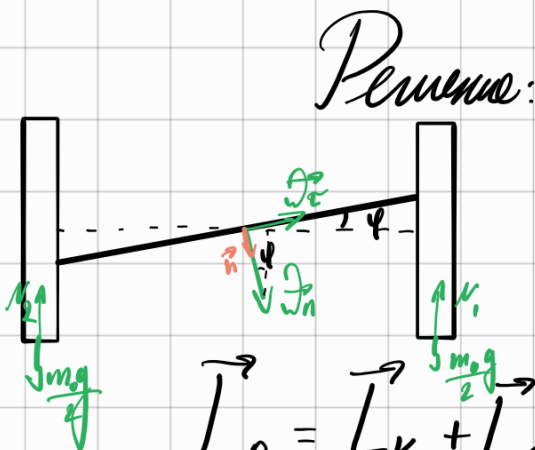
$$l = 15.1$$

$$\alpha = 1^\circ$$

$$\omega = 50 \text{ рад/с}$$

$$m_0 = 1000 \text{ кг}$$

Найти: $N_{\max}$



$$\vec{L}_0 = \vec{L}_K + \vec{L}_{CT}$$

$$\vec{M} = [\vec{\omega} \times \vec{L}] = [\vec{\omega} \times \vec{L}_K] + [\vec{\omega} \times \vec{L}_{CT}]$$

$$\vec{L}_{CT} = I_n \vec{\omega}_n + I_c \vec{\omega}_c = m \frac{l^2}{12} \omega \sin \varphi \cdot \vec{n}$$

$$M = (N_2 - N_1) \frac{l}{2} \cos \varphi = \frac{m l^2 \omega^2 \sin \varphi}{12} \cdot \cos \varphi$$

$$N_1 + N_2 = m_0 g$$

$$N_2 - N_1 = \frac{m l \omega^2 \sin \varphi}{6}$$

$$N_1 + N_2 = m_0 g$$

$$N_2 = \frac{m l \omega^2 \sin \varphi}{12} + \frac{m_0 g}{2}$$

$$N_{\max} \Rightarrow \sin \varphi \Rightarrow \max \Rightarrow \varphi = \alpha$$

$$N_{\max} = \frac{m \omega^2 l \sin \alpha}{12} + \frac{m_0 g}{2} = 6091 \text{ Н}$$

Ответ: 6091 Н

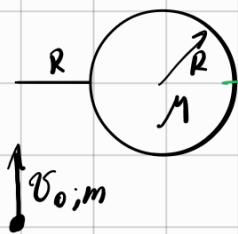
11.20

Дано:

$$\frac{R \omega}{v_0} = 5.0$$

Найти: φ

Решение



$$ЗЧУ: m v_0 = M v$$

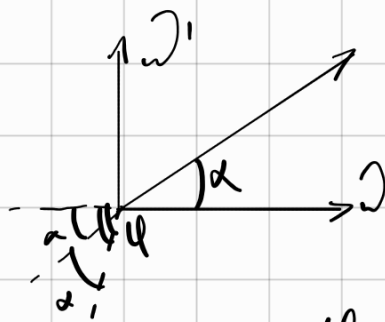
$$ЗЧУ: m v_0 \cdot 2R = \frac{2}{5} M R^2 \omega$$

$$ЗЧ: \frac{m v_0^2}{2} + \frac{2}{5} \frac{M R^2 \omega^2}{2} = \frac{M v^2}{2} + \frac{2}{5} M R^2 \frac{\omega^2 R^2}{2}$$

$$\alpha = \arctg \frac{\omega_1}{v}$$

$$v = \frac{m v_0}{M}$$

$$\omega_1 = 5 \frac{m v_0}{M R}$$



угол между резулт.
вектором бразента α ,
здравим max отклонение

$$\varphi = 2\alpha$$

$$\frac{m v_0^2}{2} + \frac{2}{5} \frac{M R^2}{2} \omega^2 = \frac{M v^2}{2} + \frac{2}{5} M R^2 \frac{(\omega^2 + \omega_1^2)}{2}$$

$$m v_0^2 + \frac{2}{5} M R^2 \omega^2 = \frac{m^2}{M} v_0^2 + \frac{2}{5} M R^2 \left(\omega^2 + 25 \frac{m^2}{M^2} \frac{v_0^2}{R^2} \right)$$

$$\frac{m}{M} v_0^2 = \frac{m^2}{M^2} v_0^2 + \frac{2}{5} R^2 \cdot 25 \frac{m^2}{M^2} \cdot \frac{v_0^2}{R^2}$$

$$1 = \frac{m}{M} + 10 \frac{m}{M} = 11 \frac{m}{M} \quad \frac{m}{M} = \frac{1}{11}$$

$$\varphi = 2 \arctg \frac{5 m v_0}{M R \omega} = 2 \arctg \frac{5}{11 \cdot 50} = 2 \arctg \frac{1}{110} \approx 1^\circ$$

Ответом 1°

XI Глава II уровень

5.49

Решение:

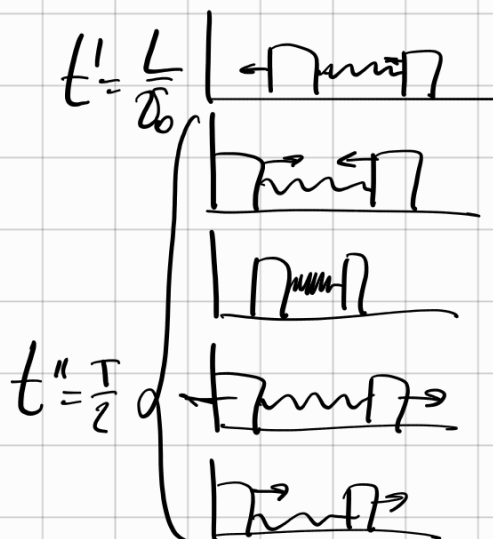
Дано:
 $v_0; m; K; L$

Найти: t

$$T = 2\pi \sqrt{\frac{m'}{K}}$$

$$m' = \frac{m \cdot m}{m+m} = \frac{m}{2}$$

$$t = \frac{2L}{v_0} + \pi \sqrt{\frac{m'}{2K}}$$



$$t''' = \frac{L}{v_0} \quad | \quad \text{spring compressed}$$

Ответ: $\frac{2L}{v_0} + \pi \sqrt{\frac{m}{2K}}$

10.78

Дано:

a

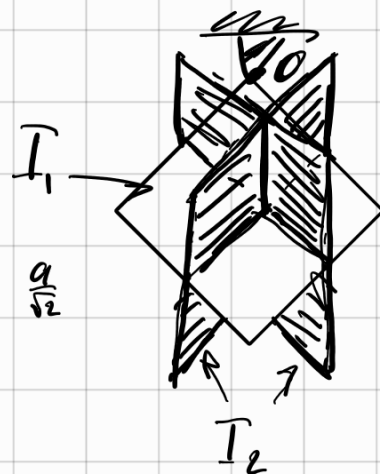
Найти: T

Решение

$$T = 2\pi \sqrt{\frac{I}{mgl}}$$

$$m = 3m_0$$

$$l = \frac{a \cdot \sqrt{2}}{2} = \frac{a}{\sqrt{2}}$$



отн. О.

$$\begin{cases} I = I_1 + 2I_2 \\ I_1 = \frac{m_0 a^2}{6} + \frac{m a^2}{2} = \frac{2}{3} m_0 a^2 \\ I_2 = \frac{m_0 a^2}{12} + \frac{m_0 a^2}{2} = \frac{7}{12} m_0 a^2 \end{cases}$$

$$I = \frac{11}{6} m_0 a^2$$

$$T = 2\pi \sqrt{\frac{11\sqrt{2} m_0 a^2}{6 \cdot 3m_0 \cdot g \cdot a}} = 2\pi \sqrt{\frac{11\sqrt{2} a}{18g}}$$

Ответ: $2\pi \sqrt{\frac{11\sqrt{2} a}{18g}}$

10.43

Дано:

$l; m; R; M$

Найти: T

Решение:

$$T = 2\pi \sqrt{\frac{I}{m g r}}$$

$$m_0 = m + M$$

$$r = \frac{m \cdot \frac{l}{2} + M \cdot l}{m + M}$$

$$I = \frac{m l^2}{12} + \frac{m l^2}{4} + m l^2 = \frac{m l^2}{3} + m l^2$$

$$T = 2\pi \sqrt{\frac{(\frac{m l^2}{3} + m l^2)(1+m)}{(m \frac{l}{2} + m l)(m+1)g}} = 2\pi \sqrt{\frac{2l}{3g} \frac{(m+3)}{(m+2)}}$$

$$\text{Окончательно: } 2\pi \sqrt{\frac{2l}{3g} \frac{(3m+m)}{(2m+m)}}$$

10.36

Дано:

$R; m; 2m; K; l;$

$$I_1 = I_2 = \frac{m R^2}{2}$$

Найти: $T; A$

Решение

$$F_{y1} + F_{TP1} = -a_1 m$$

$$F_{y2} + F_{TP2} = -a_2 2m$$

$$F_{TP1} \cdot R = I \cdot \beta_1 = I \cdot \frac{a_1}{R} = \frac{m R^2}{2} \cdot \frac{a_1}{R} = \frac{m R a_1}{2}$$

$$F_{TP1} = \frac{m a_1}{2}$$

$$F_{TP2} \cdot R = I \cdot \beta_2 = I \cdot \frac{a_2}{R} = \frac{m R^2}{2} \cdot \frac{a_2}{R} = \frac{m R a_2}{2}$$

$$F_{TP2} = \frac{m a_2}{2}$$

$$F_{y1} = 2Kx = -\frac{3}{2} m a_1$$

$$a_1 + a_2 = \ddot{x}$$

$$F_{y2} = 2Kx = -\frac{5}{2} m a_2$$

$$-\frac{4Kx}{3m} = a_1$$

$$-\frac{4Kx}{5m} = a_2$$

$$\ddot{x} = -\frac{4Kx}{m} \left(\frac{1}{3} + \frac{1}{5} \right) = -\frac{32}{15} \frac{K}{m} x$$

$$\ddot{x} + \frac{32}{15} \frac{K}{m} x = 0 \quad \omega = \sqrt{\frac{32K}{15m}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{15m}{32K}}$$

$$a_{y1} = \left(\frac{\frac{4Kx \cdot 2m}{5m} - \frac{4Kx \cdot m}{3m}}{3m} \right) = \frac{4}{45} \frac{Kx}{m}$$

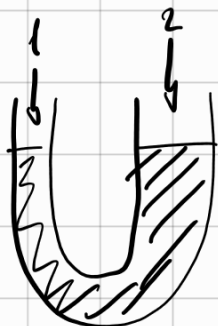
$$a = a_{y1} \cos \omega t \Rightarrow x = -\frac{a_{y1}}{\omega^2} \cos \omega t$$

$$A = \frac{a_{y1}}{\omega^2} = \frac{4kl}{45m} \cdot \frac{15m}{32k} = \frac{l}{24}$$

$$\text{Answer: } 2\pi \sqrt{\frac{15m}{32k}}; \frac{l}{24}$$

14.3

Дано:
H
Найти: T



Решение:

$$E_k = \frac{\rho \cdot S_1 \cdot H \cdot v_1^2}{2} + \frac{\rho \cdot S_2 \cdot H \cdot v_2^2}{2} \quad \begin{matrix} x_1 \cdot S_1 = x_2 \cdot S_2 \\ \downarrow \\ v_1 \cdot S_1 = v_2 \cdot S_2 \end{matrix}$$

$$E_k = \frac{\rho H}{2} \left(S_1 v_1^2 + v_1^2 \cdot \frac{S_1^2}{S_2} \right) = \frac{\rho H}{2} S_1 v_1^2 \left(1 + \frac{S_1}{S_2} \right)$$

$$\Delta E_n = \rho \cdot S_1 \cdot x_1 \cdot g \cdot \frac{x_1 \left(1 + \frac{S_1}{S_2} \right)}{2} = \frac{\rho S_1 g}{2} \left(1 + \frac{S_1}{S_2} \right) x_1^2$$

$$\frac{\rho H}{2} S_1 v_1^2 \left(1 + \frac{S_1}{S_2} \right) + \frac{\rho S_1 g}{2} \left(1 + \frac{S_1}{S_2} \right) x_1^2 = E \quad \left| \frac{d}{dt} \right.$$

$$\frac{\rho S_1}{2} \left(1 + \frac{S_1}{S_2} \right) \left(H \cdot \frac{v_1^2}{dt} + g \cdot \frac{x_1^2}{dt} \right) = 0$$

$$H \cdot a_1 \cdot v_1 + g \cdot v_1 \cdot x_1 = 0$$

$$a_1 + \frac{g}{H} x_1 = 0 \quad \omega = \sqrt{\frac{g}{H}} \quad T = 2\pi \sqrt{\frac{H}{g}}$$

$$\text{Answer: } 2\pi \sqrt{\frac{H}{g}}$$

XII Реген

II problem

Решение

12.82

Дано:

$$\omega = 3,1 \text{ c}^{-1}$$

$$L = 1,75 \text{ м}$$

Найти: N, φ, α

B CO уравна:

$$M_1 = mg \cdot \frac{L}{2} \cdot \sin \alpha$$

$$dM_2 = dF_y \cdot x \cdot \cos \alpha$$

$$dF_y \cdot x = dm \cdot \omega^2 \cdot x \cdot \sin \alpha$$

$$dm = \frac{m \cdot dx}{L}$$

$$dF_y \cdot x = \frac{\omega^2 m \sin \alpha}{L} x^2 dx$$

$$F_y \cdot x = \int_0^L \frac{\omega^2 m \sin \alpha}{L} x^2 dx = \frac{\omega^2 m \sin \alpha L}{2}$$

$$M_2 = \int_0^L \frac{\omega^2 m \sin \alpha \cos \alpha}{L} x^2 dx = \frac{\omega^2 m \sin \alpha \cos \alpha L^2}{3}$$

$$\frac{mgL \sin \alpha}{2} = \frac{\omega^2 m \sin \alpha \cos \alpha L^2}{3}$$

$$\frac{g}{2} = \frac{\omega^2 \cos \alpha L}{3}$$

$$\alpha = \arccos \frac{3g}{2\omega^2 L} \approx 29^\circ$$

$$O_x: F_y \cdot x = N \cdot \sin \varphi$$

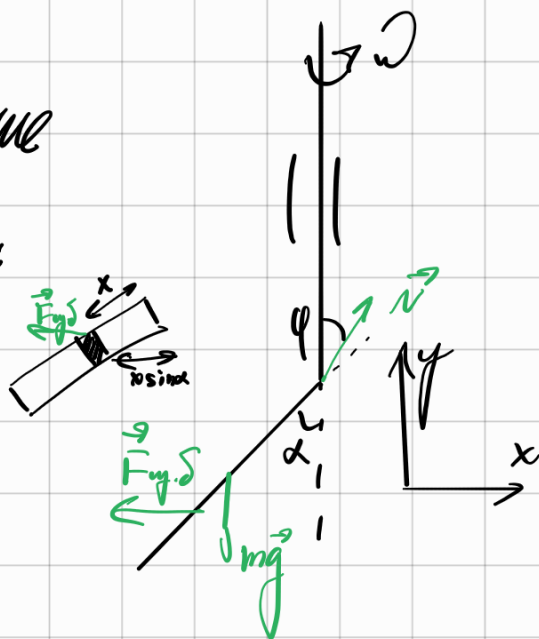
$$O_y: mg = N \cos \varphi$$

$$\frac{\omega^2 m \sin \alpha L}{2} = \tan \varphi \cdot mg$$

$$\varphi = \arctg \frac{\omega^2 \sin \alpha L}{2g} = 22^\circ$$

$$N = \frac{mg}{\cos \varphi} = 1,1 mg$$

Ответ: $\alpha = 29^\circ$
 $\varphi = 22^\circ$
 $N = 1,1 mg$



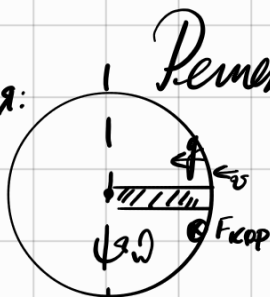
N 12.70

Дано:

Земля

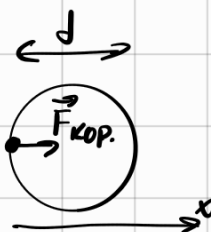
Найти: $\downarrow$

Земля:



Решение.

Цикла



$$a_r = -\frac{1}{3} \pi G \rho r = -\frac{g}{2} \cdot r \Rightarrow \omega_r = \sqrt{\frac{g}{2}}; r = R \cos \omega_r t$$

у-е [↑] радиуса го
уменьш на время

$$v_r = -R \omega_r \sin \omega_r t$$

$$ma = F_{\text{коп}} = 2m v_r \omega \quad a = 2 R \omega_r \sin(\omega_r t) \cdot \omega$$

$$v = -2 R \cos(\omega_r t) \omega + C \quad \text{из нач. данных} \quad C = 2 R \omega$$

$$v = -2 R \cos(\omega_r t) \omega + 2 R \omega$$

$$x = -\frac{2 R \sin(\omega_r t) \omega}{\omega_r} + 2 R \omega t$$

$$l = x(t) \quad t = \frac{2\pi}{\omega_r} \cdot \frac{1}{4}$$

$$l = \frac{R \omega}{\omega_r} (\pi - 2) = R \omega \sqrt{\frac{R}{g}} (\pi - 2) \approx 423 \mu\text{m}$$

Ответ: 423 мк

12.80

Дано:

$\alpha; m; K$

Найти: T

Решение

В со уравн

$$-Kx + m \alpha^2 x + mg \cos \alpha t = m \ddot{x}$$

$$\ddot{x} + x \left(\frac{K}{m} - \alpha^2 \right) = g \cos \alpha t$$

$$x = -A \cos \alpha t$$

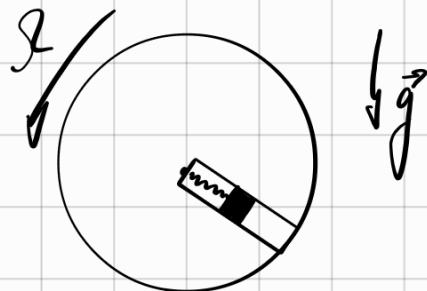
$$\ddot{x} = A \alpha^2 \cos \alpha t$$

$$A \alpha^2 \cos \alpha t - A \cos \alpha t \left(\frac{K}{m} - \alpha^2 \right) = g \cos \alpha t$$

$$A \left(\alpha^2 + \alpha^2 - \frac{K}{m} \right) = g$$

$$A = \frac{g}{2\alpha^2 - \frac{K}{m}}$$

Ответ: $\frac{g}{2\alpha^2 - \frac{K}{m}}$



12.86

Дано:

$$R = 1.6 \text{ м}$$

$$\omega = 1 \frac{\text{рад}}{\text{с}}$$

$$m = 1 \text{ кг}$$

$$K = 1 \frac{\text{Н}}{\text{м}}$$

$$l = 1 \text{ м}$$

$$v = 1 \text{ м/с}$$

Вопрос: насколько изогнется?

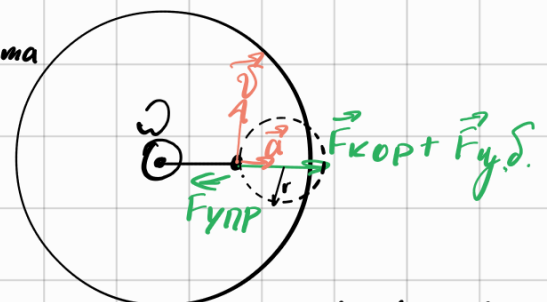
Решение

$$2m v \omega + m \omega^2 l - l \cdot K = ma$$

$$a = \frac{v^2}{r}$$

$$r = \frac{m v^2}{2m v \omega + m \omega^2 l - l K}$$

$$= \frac{1}{2 \cdot 1 + 1 - 1} = \frac{1}{2} \text{ м}$$



$$R_{\text{max}} = R + 2r = 2.1 < 1.9 = R \Rightarrow \text{выгнет}$$

Ответ: давит

XIII Задача II уровень

13.34

Решение:

Найти: $\frac{\epsilon_1}{\epsilon_2}$



$$m = \rho \cdot S(L-r)$$

$$R_{y,1} = r + \frac{L-r}{2} = \frac{L+r}{2}$$

$$a_{y,1} = \omega^2 R_{y,1}$$

$$T = m a_{y,1} = \rho S \omega^2 \frac{L^2 - r^2}{2}$$

$$\frac{\Delta l}{l} = \frac{\sigma}{E} = \frac{T}{ES}$$

$$\Delta l = \frac{\rho \omega^2 (L^2 - r^2)}{2E} \int r$$

$$\Delta l = \frac{\rho \omega^2 L^3}{3E}$$

$$\epsilon_1 = \frac{\rho \omega^2 L^2}{3E}$$

После увода у.н. стержня
каждое элементарное движение
со скоростью $\frac{L\omega}{2}$, его край
движется со скоростью $L\omega$, значит
в СО движ. стержня его ускорен
скорост будет равна: $\frac{\omega L - \frac{L\omega}{2}}{\frac{L}{2}} = \omega$

$$\epsilon_2 = \frac{\rho \omega^2 (\frac{L}{2})^2}{3E}$$

$$\boxed{\frac{\epsilon_1}{\epsilon_2} = 4}$$

Ответ: увеличится в 4 раза

13.49

Дано:

$$R = 100 \text{ м}$$

$$\omega = 400 \text{ с}^{-1}$$

$$E = 2 \cdot 10^{11} \text{ Па}$$

$$\rho = 7800 \text{ кг/м}^3$$

Найти: $T_{пр}$; ΔD

Решение:

$$\sigma = E \epsilon = E \frac{\Delta R}{R}$$

$$\sigma = \frac{F}{S} = \frac{m \omega^2 R}{\pi S} = \frac{\rho \cdot S \cdot R \omega^2 R}{\pi S} = \rho R^2 \omega^2$$

$$\Delta R = \frac{\rho R^3 \omega^2}{E}$$

$$\Rightarrow \Delta D = 2 \Delta R = \frac{8 \pi^2 \rho \omega^2 R^3}{E} = 0,49 \text{ м}$$

$$T_{пр} > \sigma = \rho R^2 4 \pi^2 \omega^2 = 493 \cdot 10^6 \text{ Па}$$

Ответ: 0,49 м
493 МПа

14.39

Дано:

$$v = 0,3 \text{ м/с}$$

$$\rho = 0,9 \text{ т/м}^3$$

$$v_{\text{ж}} = 1500 \text{ м/с}$$

Найти: P

Решение:

$$V = S \cdot v_{\text{ж}} \cdot dt$$

$$m = \rho V = \rho \cdot S \cdot v_{\text{ж}} \cdot dt$$

$$dp = m v = \rho S v_{\text{ж}} v \cdot dt$$

$$F = \frac{dp}{dt} = \rho S v_{\text{ж}} v$$

$$P = \frac{F}{S} = \rho v_{\text{ж}} v = 405000 \text{ Па} \approx 4 \text{ атм.}$$

Ответ: 4 атм.

Т 13.1

Дано:

$$\sigma_r = 100 \text{ ГПа}$$

$$\rho = 1300 \text{ кг/м}^3$$

$$R = 6380 \text{ км}$$

$$g = 9,81 \text{ м/с}^2$$

Вопрос: гостановится ли провалить нуден?

Решение:



$$a = g_0 \frac{R^2}{r^2} - \omega^2 r$$

$$d\sigma = a \rho dr$$

$$a = 0 = d\sigma = 0 \Rightarrow \sigma \Rightarrow \max$$

$$r_{\max} = \sqrt[3]{\frac{g_0 R^2}{\omega^2}}$$

$$\sigma_{\max} = \int_R^{r_{\max}} \left(g_0 \frac{R^2}{r^2} - \omega^2 r \right) \rho dr =$$

$$= \rho g_0 R^2 \left(\frac{1}{R} - \frac{1}{r_{\max}} \right) - \rho \frac{\omega^2}{2} (r_{\max}^2 - R^2) =$$

$$= 63 \text{ ГПа} < \sigma_r \Rightarrow \text{гостановится}$$

Ответ: гостановится

XIV Глава II упробн.

8.7

Дано:

$$V = 0,6c$$

$$l = \frac{L}{2}$$

$$t = 2 \text{ месяца}$$

$$\text{Найти: } \Delta t'$$

Решение

$$ct_1 = l - Vt_1 \quad t_1 = \frac{L}{2(c+V)}$$

$$ct_2 = l + Vt_2 \quad t_2 = \frac{L}{2(c-V)}$$

$$\Delta t = t_2 - t_1 = \frac{L}{2} \left(\frac{1}{c-V} - \frac{1}{c+V} \right) =$$

$$= \frac{L}{2} \left(\frac{c+V - c+V}{c^2 - V^2} \right) = \frac{LV}{c^2 - V^2} = \frac{L \cdot 0,6c}{c^2 - 0,36c^2} =$$

$$= \frac{L}{c} \cdot \frac{0,6}{0,64} = \frac{L}{c} \cdot \frac{15}{16}$$

$$\Delta t' = \Delta t \cdot \sqrt{1 - \beta^2} = \frac{L}{c} \cdot \frac{15}{16} \cdot \frac{4}{5} = \frac{3}{4} \frac{L}{c} = 1,5 \text{ месяца}$$

Ответ: 1,5 месяца

8.30

Дано:

$$V = 0,8c$$

$$\Delta t = 1c.$$

$$\text{Найти: } \Delta t'$$

Решение

$$\Delta t' = \Delta t \sqrt{\frac{1+\beta}{1-\beta}} \quad \text{— релятивистский эфф. Доплера}$$

$$\Delta t' = 1 \sqrt{\frac{1,8}{0,2}} = 3c.$$

Ответ: 3c.

8.98

Дано:

$$\pi^0; \pi^+$$

$$L = 6 \text{ км}$$

$$v_1 = 0,8c$$

$$v_2 = 0,6c$$

Решение:

$$\tau = \tau' \gamma = \frac{\tau'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\tau_1 = \frac{\tau'_1}{0,6} \quad \tau_2 = \frac{\tau'_2}{0,8}$$

$$\tau_1' = 6,7 \cdot 10^{-17} \text{ c}$$

$$\tau_2' = 2,6 \cdot 10^{-8} \text{ c}$$

$$\tau_1 = 14,5 \cdot 10^{-17} \text{ c}$$

$$\tau_2 = 3,25 \cdot 10^{-8} \text{ c}$$

Вопрос: улетит ли π^0 до детектора π^+

$$\Delta t = \frac{L}{v_1 - v_2} = \frac{L}{0,2c} = 10^{-16} \text{ c}$$

$$\left. \begin{array}{l} \Delta t < \tau_1 \\ \Delta t < \tau_2 \end{array} \right\} \Rightarrow \text{горит}$$

Ответ: горит

8.89

Дано:

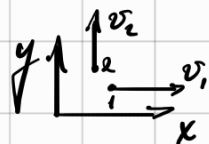
$$\beta_1 = 0,65$$

$$\beta_2 = 0,8$$

Найти: $v_{\text{отн}}$

Решение:

$$v_1 = 0,65c \quad v_2 = 0,8c$$



Переходим в С.О. 1 частицы:

$$v_{2x} = \frac{v_{2x}' + v_1}{1 + \frac{v_{2x}' v_1}{c^2}} = 0 \Rightarrow v_{2x}' = -v_1$$

$$v_{2y} = \frac{v_{2y}' \sqrt{1 - \beta_1^2}}{1 + \frac{v_{2x}' v_1}{c^2}} = v_2 \Rightarrow v_{2y}' = v_2 \sqrt{1 - \beta_1^2}$$

$$v_{\text{отн}} = \sqrt{v_1^2 + v_2^2 (1 - \beta_1^2)} = \sqrt{v_1^2 + v_2^2 - \frac{v_1^2 v_2^2}{c^2}}$$

$$= 0,89c$$

Ответ: 0,89c

XV Регла

8.44

Дано:

$\pi^+; \mu^+; \gamma$

$$m_{\pi}c^2 = 139,6 \text{ МэВ}$$

$$m_{\mu}c^2 = 105,7 \text{ МэВ}$$

$$m_{\gamma}c^2 = 0$$

Найти: $E_{\mu}; E_{\gamma}$

II уровень

Решение:

$$E_{\pi} = E_{\mu} + E_{\gamma} - 3c^2$$

$$E_{\pi} = m_{\pi}c^2$$

$$E_{\gamma} = p_{\gamma}c$$

$$E_{\mu} = \sqrt{p_{\mu}^2c^2 + m_{\mu}^2c^4}$$

$$p_{\mu} + p_{\gamma} = 0 - \text{закон}$$

$$E_{\mu} = \sqrt{E_{\gamma}^2 + m_{\mu}^2c^4}$$

$$E_{\pi} = \sqrt{E_{\gamma}^2 + m_{\mu}^2c^4} + E_{\gamma}$$

$$E_{\gamma} = 30 \text{ МэВ} \quad E_{\mu} = 110 \text{ МэВ}$$

$$E_{\mu\gamma} = E_{\gamma} = 30 \text{ МэВ}$$

$$E_{\mu\pi} = E_{\mu} - m_{\mu}c^2 = 4 \text{ МэВ} \quad \text{Ответ: } 30 \text{ МэВ}$$

$$4 \text{ МэВ}$$

8.57

Дано:

$$K^+ \rightarrow \pi^+ + \pi^0 \rightarrow \pi^+ + \gamma + e^+ + e^-$$

$$l = 0,38 \text{ нм}$$

$$E_{0K} = 494 \text{ МэВ}$$

$$E_{0\pi^+} = 140 \text{ МэВ}$$

$$E_{0\pi^0} = 135 \text{ МэВ}$$

Найти: τ_0

Решение:

$$\pi^- \leftarrow K^+ \rightarrow \pi^+ \rightarrow e^+ \rightarrow e^-$$

$$E_{0K} = E_{0\pi^+} + E_{\mu\pi^+} + E_{0\pi^0} + E_{\mu\pi^0}$$

$$E_{0\pi^+} \approx E_{0\pi^0} \Rightarrow \text{Судя считать } E_{\mu\pi^+} \approx E_{\mu\pi^0}$$

$$E_{\mu\pi} = \frac{E_{0K} - E_{0\pi^+} - E_{0\pi^0}}{2} = 109,5 \text{ МэВ}$$

$$E_{\pi^0} = E_{0\pi^0} + E_{\mu\pi} = 244,5 \text{ МэВ}$$

$$E_{\pi^0} = E_{0\pi^0} \gamma = \frac{E_{0\pi^0}}{\sqrt{1-\beta^2}} \quad \beta = \sqrt{1 - \left(\frac{E_{0\pi^0}}{E_{\pi^0}}\right)^2} = 0,834$$

$$\text{В ЛСО } \tau = \frac{l}{\beta c} = 1,52 \cdot 10^{-15} \text{ с}$$

$$\text{В СО } \tau_0 = \tau \sqrt{1-\beta^2} = 8,4 \cdot 10^{-16} \text{ с}$$

$$\text{Ответ: } 8,4 \cdot 10^{-16} \text{ с}$$

T 15.3

Решение:

Law:

$$l = 10 \text{ cm}$$

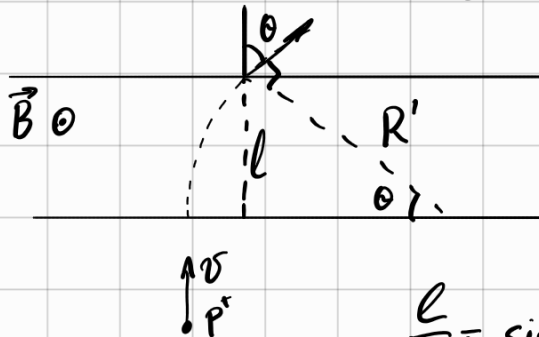
$$B = 1 \text{ T}$$

$$E = g, u \Gamma \partial B$$

$$m = 0,94 \text{ TgB/Cu}$$

$$\frac{q}{m} = 9,6 \cdot 10^7 \frac{V_1}{V_2}$$

Hamm: @



$$\frac{L}{R} = \sin \theta \approx \theta$$

$$F = qvB \cdot \cos \alpha$$

$$q_c' B = m \frac{c_L}{R}$$

$$R = \frac{m c}{q B} - b \cos p^+$$

$$R' = R_X - b \cdot \Delta CO$$

$$\Theta = \frac{q_B \ell}{mc \gamma} = 3,2 \cdot 10^{-3} \text{ rad}$$

$$\gamma = \frac{E}{mc^2} = 10$$

Ombem: $3,2 \cdot 10^{-3}$ pag.

8.105

Dano:

$$V = 2,6 \cdot 10^8 \text{ m/s}$$

$$E_k' = \frac{1}{2} E_k$$

Stamm: m_x

$$\gamma_1 = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 2$$

$$\frac{j_1 - 1}{j_2 - 1} = 2$$

$$f_2 = 1,5$$

Revenue:

$$E = \gamma E_0$$

$$E_k = E - E_0 = (j-1)E_0$$

ЗСУ:

$$m_x^2 c^2 (y_x^2 - 1) = m_n^2 c^2 (y_1^2 - 1) + m_n^2 c^2 (y_2^2 - 1)$$

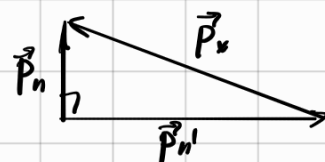
$$m_x^2 (f_x^2 - 1) = (f_1^2 + f_2^2 - 2) m_n^2$$

300:

$$m_n f_1 c^2 + m_x c^2 = m_n f_2 c^2 + m_x f_x c^2$$

$$m_x(1 - f_x) = m_n(f_2 - f_1)$$

$$f_x = \frac{m_n}{m_x} (\delta_1 - \delta_2) + 1$$



$$m_x^2 \left(\frac{m_n^2 (x_1 - x_2)^2}{m_x^2} + \frac{2m_n}{m_x} (x_1 - x_2) \right) = (x_1^2 + x_2^2 - 2) m_n^2$$

$$m_n \cdot 0,25 + m_x = 4,25 m_n$$

$$m_x = 4 m_n$$

Answer: $4 m_n$